

Why Every Dynamic System Presupposes a Predicate

Toward a General Theory of Admissibility

Abstract

Most scientific theories focus on dynamics: how systems evolve, compute, adapt, or transform over time. Far less attention has been given to a logically prior question: **what determines which states are admissible before any dynamics can occur?**

This paper proposes that every dynamic system presupposes an **admissibility predicate**: a logical structure defining the subset of states upon which the system's dynamics are meaningful. Dynamics do not generate this predicate; rather, they operate within it. This distinction unifies concepts traditionally viewed as unrelated: axioms in formal logic, boundary conditions in engineering, fitness constraints in biology, loss functions in machine learning, and price systems in economics. The paper introduces a formal definition of an admissibility predicate and argues that the dependence of dynamics upon such predicates represents a universal structural principle rather than a domain-specific observation.

1. Introduction

Scientific explanation usually begins with evolution equations.

Physics begins with differential equations.

Biology begins with evolution.

Economics begins with exchanges.

Artificial intelligence begins with optimization.

Logic begins with inference.

Yet all these theories quietly assume something that almost never becomes the object of analysis:

Which states are even allowed to participate in the dynamics?

Before asking how a system evolves, one must first specify the domain in which evolution is defined.

This paper argues that this domain is determined by what we call an **admissibility predicate**.

The central thesis is:

Every dynamic system presupposes a predicate defining its admissible state space. Dynamics cannot generate that predicate because every transition is evaluated relative to it.

2. Dynamics Require Domains

Consider a differential equation

$$\frac{dx}{dt} = F(x).$$

The equation itself does not determine

- the initial conditions,
- the boundary conditions,
- the admissible domain,
- or whether solutions even exist.

Exactly the same observation appears elsewhere.

A neural network minimizes a loss function.

But the optimization does not create the loss function.

Evolution selects among viable organisms.

It does not create viability itself.

Formal deduction derives theorems.

It does not derive every axiom upon which deduction depends.

Thus, dynamics appear universally subordinate to a prior logical structure.

3. Definition: Admissibility Predicate

Let

$$S$$

be the universe of possible descriptions or states.

An **admissibility predicate** is a mapping

$$P : S \rightarrow \{0, 1\}$$

such that

$$P(s) = 1$$

iff state

$$s \in S$$

belongs to the admissible domain.

The admissible region is therefore

$$R = \{s \in S \mid P(s) = 1\}$$

Dynamics are defined only over

$$R.$$

Consequently,

$$D : R \rightarrow R.$$

The predicate does not describe evolution.

It defines the logical possibility of evolution.

4. Fundamental Principle

Principle of Logical Priority

Let

$$D$$

denote any operational dynamics.

Then

$$D$$

presupposes an admissibility predicate

$$P.$$

Without

$$P,$$

the dynamics possess no well-defined operational domain.

Equivalently,

$P \prec D$

where

$$\prec$$

denotes logical precedence rather than temporal precedence.

This statement should not be interpreted causally.

It expresses a condition of formulation.

One cannot define dynamics before specifying the admissible domain on which those dynamics act.

5. Why Dynamics Cannot Generate Their Own Predicate

Suppose that

$$P$$

were generated entirely by

$$D.$$

Then evaluating whether a transition is admissible would require executing

$$D.$$

But executing

$$D$$

already presupposes that the current state belongs to its domain.

Therefore,

$$P$$

must already exist before

$$D$$

can operate.

Otherwise the dynamics become undefined.

Hence

$$P \notin \text{Closure}(D).$$

The admissibility predicate cannot be generated solely by the operational closure of the dynamics it defines.

This resembles—but is more general than—the familiar impossibility results encountered in logic, computation, and control theory.

6. Examples Across Disciplines

Logic

Predicate:

Consistency under axioms.

Dynamics:

Formal derivation.

Without axioms, inference has no admissible starting point.

Engineering

Predicate:

Boundary conditions.

Dynamics:

Differential equations.

Without supports or constraints, equations describe infinitely many unrealizable solutions.

Evolution

Predicate:

Viability.

Dynamics:

Natural selection.

Selection operates only among organisms already capable of reproduction.

Artificial Intelligence

Predicate:

Loss function.

Dynamics:

Gradient descent.

Learning minimizes error relative to a previously defined objective.

The optimization process does not generate its own objective.

Economics

Predicate:

Informational admissibility expressed through prices.

Dynamics:

Production, exchange, and allocation.

Prices constrain the admissible space of economic decisions.

Removing them leaves dynamics without coherent optimization criteria.

7. Relation to the Theory of Axiomatic Necessity

Within TNA,

$$N_0$$

represents the operational domain.

The admissibility predicate belongs to

$$N_1.$$

Operational dynamics evolve within

$$N_0,$$

but admissibility is determined relative to

$$N_1.$$

The structural relation therefore becomes

$$N_0 \not\prec N_1.$$

This is not merely a theorem about formal logic.

It expresses a universal distinction between operational evolution and the logical conditions that render that evolution meaningful.

8. Admissibility and Structural Incoherence

Let

$$I_s$$

measure structural incoherence.

Then

$$I_s = 0$$

corresponds to complete compatibility between system states and the admissibility predicate.

As systems evolve,

$$I_s$$

measures their distance from admissibility.

Learning, scientific progress, engineering optimization, and biological adaptation may all be interpreted as processes that reduce

$$I_s$$

within a fixed admissible domain.

The dynamics optimize.

The predicate defines what optimization means.

9. Philosophical Consequences

The proposed principle shifts the emphasis of scientific explanation.

Instead of asking only

What causes change?

we must also ask

What determines which changes are even admissible?

This distinction separates dynamics from possibility.

Most scientific theories investigate the former.

The latter has received comparatively little formal treatment despite preceding every dynamic formulation.

10. Conclusion

The central claim of this paper is simple.

Dynamics never begin from nothing.

Every dynamic system presupposes a predicate defining its admissible state space.

This predicate is logically prior to dynamics because every transition is evaluated relative to it.

The principle applies uniformly across logic, engineering, biology, artificial intelligence, economics, and any discipline whose systems evolve under constraints.

The consequence is profound.

The deepest structure of scientific explanation may not be dynamics themselves, but the predicates that make dynamics possible.

Admissibility precedes dynamics because possibility precedes evolution.

References

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